

Homework 2

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March 2024

Exercise 1

a.

By Feynman-Kac Theorem,

$$V(t, X) = \mathbb{E}^{\mathbb{Q}} [\log X(T)^3 | \mathcal{F}(t)]$$

for any $t \leq T$ where $dX(t) = \mu X(t)dt + \sigma X(t)dW^{\mathbb{Q}}(t)$. Since $X(t)$ follows the GBM, we obtain, by (2.17), that

$$X(T) = X(t) \exp \left(\left(\mu - \frac{\sigma^2}{2} \right) (T - t) + \sigma (W^{\mathbb{Q}}(T) - W^{\mathbb{Q}}(t)) \right).$$

Hence,

$$\begin{aligned} \log X(T)^3 &= 3 \log X(T) \\ &= 3 \log X(t) + 3 \left(\mu - \frac{\sigma^2}{2} \right) (T - t) + 3\sigma (W^{\mathbb{Q}}(T) - W^{\mathbb{Q}}(t)) \end{aligned}$$

Since $W^{\mathbb{Q}}(t)$ is a martingale,

$$\begin{aligned} V(t, X) &= \mathbb{E}^{\mathbb{Q}} [\log X(T)^3 | \mathcal{F}(t)] \\ &= 3 \log X(t) + 3 \left(\mu - \frac{\sigma^2}{2} \right) (T - t) + 3\sigma (W^{\mathbb{Q}}(t) - W^{\mathbb{Q}}(t)) \\ &= 3 \log X(t) + 3 \left(\mu - \frac{\sigma^2}{2} \right) (T - t). \end{aligned}$$

b.

Consider

$$\frac{V(t, S)}{M(t)} = e^{-r(t-t_0)} V(t, S).$$

We obtain dynamics

$$d\left(e^{-r(t-t_0)}V(t, S)\right) = V(t, S)d\left(e^{-r(t-t_0)}\right) + e^{-r(t-t_0)}dV(t, S). \quad (1)$$

Applying Itô's Lemma to $g(t, S) = V(t, S)$ yields that

$$dV = \left(\frac{\partial V}{\partial t} + rS\frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2}\right)dt + \sigma S\frac{\partial V}{\partial S}dW^{\mathbb{Q}}. \quad (2)$$

Substitute identity

$$d\left(e^{-r(t-t_0)}\right) = -re^{-r(t-t_0)}dt$$

to Equation (1) and multiply both sides by $e^{r(t-t_0)}$. We obtain

$$\begin{aligned} e^{r(t-t_0)}d\left(e^{-r(t-t_0)}V\right) &= e^{r(t-t_0)}V(-r)e^{-r(t-t_0)}dt + e^{r(t-t_0)}e^{-r(t-t_0)}dV \\ &= -rVdt + dV. \end{aligned}$$

Substitute Equation (2) and obtain

$$\begin{aligned} e^{r(t-t_0)}d\left(e^{-r(t-t_0)}V\right) &= -rVdt + \left(\frac{\partial V}{\partial t} + rS\frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2}\right)dt \\ &\quad + \sigma S\frac{\partial V}{\partial S}dW^{\mathbb{Q}} \\ &= \left(\frac{\partial V}{\partial t} + rS\frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - rV\right)dt \\ &\quad + \sigma S\frac{\partial V}{\partial S}dW^{\mathbb{Q}}. \end{aligned}$$

Recall that the first term is given to be zero. Hence,

$$e^{r(t-t_0)}d\left(e^{-r(t-t_0)}V\right) = \sigma S\frac{\partial V}{\partial S}dW^{\mathbb{Q}}$$

and

$$d\left(e^{-r(t-t_0)}V\right) = e^{-r(t-t_0)}\sigma S\frac{\partial V}{\partial S}dW^{\mathbb{Q}}.$$

Integrating both sides yields

$$\begin{aligned} \int_{t_0}^T e^{-r(t-t_0)}\sigma S\frac{\partial V}{\partial S}dW^{\mathbb{Q}}(t) &= \int_{t_0}^T d\left(e^{-r(t-t_0)}V\right) \\ &= e^{-r(T-t_0)}V(T, S) - V(t_0, S). \end{aligned}$$

Reshuffling gives

$$V(t_0, S) = e^{-r(T-t_0)}V(T, S) - \int_{t_0}^T e^{-r(t-t_0)}\sigma S\frac{\partial V}{\partial S}dW^{\mathbb{Q}}(t).$$

Take expectation to obtain

$$\begin{aligned}\mathbb{E}^{\mathbb{Q}} \left[V(t_0, S) \middle| \mathcal{F}(t_0) \right] &= \mathbb{E}^{\mathbb{Q}} \left[e^{-r(T-t_0)} V(T, S) \middle| \mathcal{F}(t_0) \right] \\ &\quad - \mathbb{E}^{\mathbb{Q}} \left[\int_{t_0}^T e^{-r(t-t_0)} \sigma S \frac{\partial V}{\partial S} dW^{\mathbb{Q}}(t) \middle| \mathcal{F}(t_0) \right].\end{aligned}$$

Notice that $e^{-r(t-t_0)} \sigma S \frac{\partial V}{\partial S}$ is square-integrable because

$$\begin{aligned}\mathbb{E} \left[\int_{t_0}^T e^{-2r(t-t_0)} \sigma^2 S^2 \left(\frac{\partial V}{\partial S} \right)^2 dt \right] \\ &= \sigma^2 \mathbb{E} \left[\int_{t_0}^T e^{-2r(t-t_0)} S_0^2 \exp \left((2r - \sigma^2)(t - t_0) + 2\sigma(W^{\mathbb{Q}}(t) + W^{\mathbb{Q}}(t_0)) \right) \left(\frac{\partial V}{\partial S} \right)^2 dt \right] \\ &= \sigma^2 S_0^2 \mathbb{E} \left[\int_{t_0}^T e^{-\sigma^2(t-t_0)} \exp \left(2\sigma(W^{\mathbb{Q}}(t) + W^{\mathbb{Q}}(t_0)) \right) \left(\frac{\partial V}{\partial S} \right)^2 dt \right] \\ &\leq \sigma^2 S_0^2 \mathbb{E} \left[\int_{t_0}^T e^{2\sigma(W^{\mathbb{Q}}(t) + W^{\mathbb{Q}}(t_0))} \left(\frac{\partial V}{\partial S} \right)^2 dt \right] < \infty\end{aligned}$$

because the derivative of the option is given to be square-integrable. Hence, we have a martingale Itô integral

$$I(T) := \int_{t_0}^T e^{-r(t-t_0)} \sigma S \frac{\partial V}{\partial S} dW^{\mathbb{Q}}(t).$$

Therefore,

$$\mathbb{E}^{\mathbb{Q}} \left[\int_{t_0}^T e^{-r(t-t_0)} \sigma S \frac{\partial V}{\partial S} dW^{\mathbb{Q}}(t) \middle| \mathcal{F}(t_0) \right] = \int_{t_0}^{t_0} e^{-r(t-t_0)} \sigma S \frac{\partial V}{\partial S} dW^{\mathbb{Q}}(t) = 0.$$

Consequently,

$$\mathbb{E}^{\mathbb{Q}} \left[V(t_0, S) \middle| \mathcal{F}(t_0) \right] = \mathbb{E}^{\mathbb{Q}} \left[e^{-r(T-t_0)} V(T, S) \middle| \mathcal{F}(t_0) \right]$$

and

$$V(t_0, S) = e^{-r(T-t_0)} \mathbb{E}^{\mathbb{Q}} \left[V(T, S) \middle| \mathcal{F}(t_0) \right],$$

as required.

Exercise 2

Figure 1 shows convergence of Monte Carlo simulations of the call option price as the number of generated paths increases. The standardization seems to improve the convergence because its estimates oscilate less around the theoretical value 0.998.

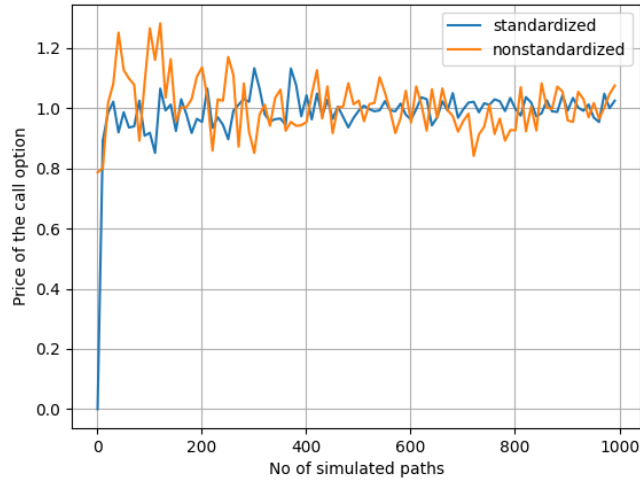


Figure 1: Convergence with and without standardization

Exercise 3

a.

The iterations of each of the three methods is shown in Figure 2 and 3 below. Notice that the iterations of the Newton-Rhapson and the Combined root-finding methods are overlapping.

b.

I acquired option market data of SAP from *yfinance* library. Figure 4 shows the volatility smile and skew as required.

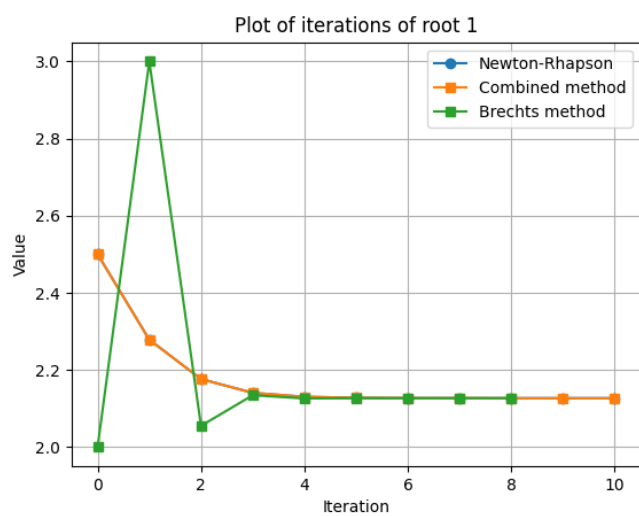


Figure 2: Root 1

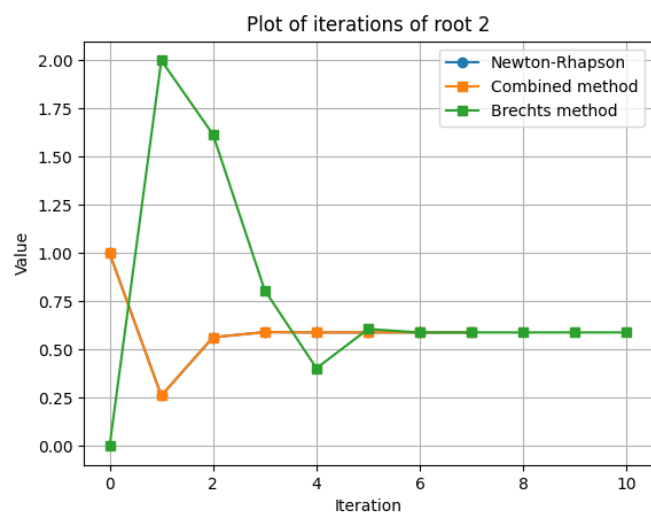


Figure 3: Root 2

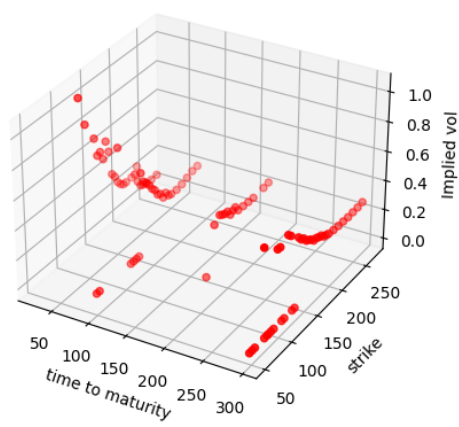


Figure 4: Volatility smile and skew